AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Midterm Examination (MTA) Makeup
Spring Semester 2024/2025

Date: 24/April/2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 90 Minutes

Instructions:

- Total Marks: **30**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple Choice Questions: [5 marks]

Question 1: Choose the most correct answer.

1. What is the role of an operating system?
   a) To provide an interface between the user and the hardware
   b) To manage the resources of a computer system
   c) To provide a platform for software applications to run
   d) All the above

2. In Client-Server Systems, which mechanism allows processes to communicate and synchronize with each other?
   a) Shared memory
   b) Message passing
   c) Sockets
   d) Process management

3..... is copying information into a faster storage system:
   a) Reading
   b) Calling
   c) Caching
   d) None of the above

4 Which of the following is a type of system call?
   a) File manipulation
   b) Memory allocation
   c) Process control
   d) All the above

5. The one program always running on the computer is:
   a) Kernel
   b) Application programs
   c) System programs
   d) All the above.

Part 2: Short Essay Questions: [15 marks]

Question 2: [2 marks]
List the four components of the computer systems.

Question 3: [2 marks]
Differentiate between data parallelism and task parallelism.

Question 4: [3 marks]
What do the following acronyms stand for?
   a) CLI
   b) PCB
   c) SAN

Question 5: [4 marks]
List three advantages of multiprogramming and talk about one of them.

Question 6: [4 marks]
   a) What are the different process states? (2.5 marks)
   b) Draw process state diagram. (1.5 marks)

Part 3: Problem Solving: [10 marks]

Question 7: [4 marks]
   a) What is the meaning of the following crontab time format? (2 marks)

8 7-10 * * *

   b) What does ***chmod*** do? Provide an example. (2 marks)

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Question 8: **[6 marks]**

Write a Bash script that prompts the user to enter **three numbers** one by one (using a loop). The script should then determine and display the **largest number** among the three entered numbers.

Expected Output (Example Run):

Enter number 1: 8
Enter number 2: 15
Enter number 3: 12
The largest number is 15.

End of Question

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